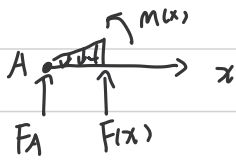


5-2.

先求支座反力

$$\begin{cases} F_A + F_B = \frac{1}{2} q_0 L \\ F_B l = \frac{1}{2} q_0 l \cdot \frac{2}{3} l \end{cases} \quad \begin{cases} F_A = \frac{1}{6} q_0 l \\ F_B = \frac{1}{3} q_0 l \end{cases}$$

求弯矩分布



$$F(x) = \frac{1}{2} \cdot \frac{q_0}{l} \cdot q_0 \cdot x - \frac{1}{6} q_0 l = \frac{1}{2} \frac{q_0}{l} x^2 - \frac{1}{6} q_0 l$$

$$\sum M_A = \frac{q_0}{2} \cdot \frac{x}{l} \cdot \frac{q_0}{2} - (\frac{1}{2} \frac{q_0}{l} x^2 - \frac{1}{6} q_0 l) x - M(x) = 0$$

$$\therefore M(x) = -\frac{1}{6} \frac{q_0}{l} x^3 + \frac{1}{6} q_0 l x$$

规定 w 向下为正

$$EI w'' = -M(x)$$

积分一次: $EI \theta(x) = EI w' = -\frac{q_0 l}{12} x^2 + \frac{q_0}{24 l} x^4 + C_1$

积分两次: $EI w(x) = -\frac{q_0 l}{36} x^3 + \frac{q_0}{120 l} x^5 + C_1 x + C_2$

边界条件: A处: $x=0, w=0 \Rightarrow C_2 = 0$
 B处: $x=l, w=0 \Rightarrow C_1 = \frac{7 q_0 l^3}{360}$

$$\begin{cases} \theta_A = \theta(0) = \frac{C_1}{EI} = \frac{7 q_0 l^3}{360 EI} \\ \theta_B = \theta(l) = -\frac{q_0 l^3}{45 EI} \end{cases}$$

令 $w'(x) = 0$ 得: $15 x^4 - 30 l^2 x^2 + 7 l^4 = 0$

$$x_0 = \sqrt{1 - \frac{2\sqrt{5}}{15}} l \approx 0.519 l$$

代入 $w(x_0) \approx 0.0065 \frac{q_0 l^4}{EI}$

5-3.

先求支座反力

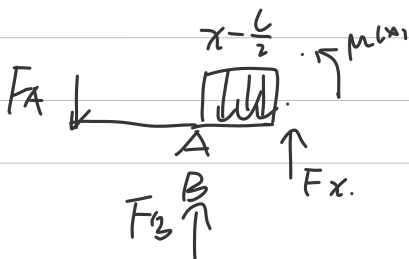
$$\begin{cases} F_A + q l = F_B + F_C \\ q l \cdot \frac{1}{2} = F_A \cdot \frac{1}{2} + l \cdot F_C \end{cases} \quad \begin{cases} F_B = \frac{5}{4} q l \\ F_C = \frac{1}{4} q l \end{cases}$$

以A为取点, 向右为正方向

当 $x \in (0, \frac{1}{2})$ 时: $M(x) = -\frac{1}{2} q l \cdot x$

当 $x \in (\frac{1}{2}, \frac{3}{2})$ 时: $\sum M_B = M(x) + (x - \frac{1}{4}) q \cdot (x - \frac{1}{2}) + \frac{1}{2} q l \cdot \frac{1}{2} l$

$$- (x - \frac{1}{2}) q \cdot \frac{1}{2} (x - \frac{1}{2}) = 0$$



$$\therefore M(x) = -\frac{1}{2} q x^2 + \frac{5}{4} q l \cdot x - \frac{3}{4} q l^2$$

积分一次: $EI\theta(x) = \frac{1}{4}flx^2 + C_1$

积分两次: $EIw(x) = \frac{1}{12}flx^3 + C_1x + C_2$ $w(\frac{1}{2}l) = 0$

积分一次: $EI\theta(x) = \frac{1}{6}fx^3 - \frac{5}{8}flx^2 + \frac{3}{4}fl^2x + C_1'$

积分两次: $EIw(x) = \frac{1}{24}fx^4 - \frac{5}{24}flx^3 + \frac{3}{8}fl^2x^2 + C_1'x + C_2'$ $w(\frac{3}{2}l) = 0$

由边界条件与连续性, 可得.

$$\theta_1(x) = \frac{1}{EI} \left(\frac{1}{4}flx^2 - \frac{5}{48}fl^3 \right)$$

$$w_1(x) = \frac{1}{EI} \left(\frac{1}{12}flx^3 - \frac{5}{48}fl^3x + \frac{1}{24}fl^4 \right) \quad (0 \leq x \leq \frac{1}{2}l)$$

$$\theta_2(x) = \frac{1}{EI} \left(\frac{1}{6}fx^3 - \frac{5}{8}flx^2 + \frac{3}{4}fl^2x - \frac{9}{32}fl^3 \right)$$

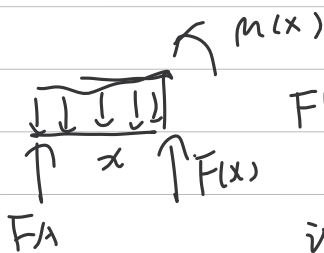
$$w_2(x) = \frac{1}{EI} \left(\frac{1}{24}fx^4 - \frac{5}{24}flx^3 + \frac{3}{8}fl^2x^2 - \frac{9}{32}fl^3x + \frac{9}{128}fl^4 \right) \quad (l/2 \leq x \leq 3/2l)$$

$$\therefore \theta_A = \theta(0) = -\frac{5}{48} \frac{fl^3}{EI} \quad w_A = \frac{fl^4}{24EI}$$

$$\theta_B = \theta(l/2) = -\frac{9}{24EI} \quad w_B = -\frac{9}{384EI} fl^4$$

$$F_A = F_B = F_C$$

$$F_A + F_B + F_C = fl \quad \therefore F_A = F_B = F_C = \frac{1}{3}fl$$



$$F(x) = flx - \frac{1}{3}fl$$

$$m(x) + (flx - \frac{1}{3}fl)x - flx \cdot \frac{x}{2} = 0$$

计算得到 $m(x) = -\frac{1}{2}fx^2 + \frac{1}{3}flx$

$$EI\theta(x) = \frac{1}{6}fx^3 - \frac{1}{6}flx^2 + C_1$$

$$EIw(x) = \frac{1}{24}fx^4 - \frac{1}{18}flx^3 + C_1x + C_2$$

$$w(0) = 0 \Rightarrow C_2 = 0 \quad \theta(l/2) = 0 \Rightarrow C_1 = \frac{fl^3}{48}$$

$$\therefore w(x) = \frac{1}{EI} \left(\frac{1}{24}fx^4 - \frac{1}{18}flx^3 + \frac{fl^3}{48}x \right)$$

$$\therefore \Delta = w(l/2) = \frac{7}{1152} \frac{fl^4}{EI}$$

